Agentic-AI project report

Topic:- Social Media Strategy Planner — A Multi-Agent AI System for Automated Campaign Intelligence

Group Members:-

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Github:- https://github.com/Tanishqbhongade/Ai-News-Research-agent/tree/main/AI-News-Research-Agentai project

Live Demo:- **https://ai-news-research-agent-27kemjahtm3ecmyxijcyry.streamlit.app/**

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1. Problem Statement

PROBLEM In today's fast-paced digital era, professionals, researchers, and enthusiasts suffer from severe information overload. When researching a specific topic or tracking breaking news, individuals must manually sift through dozens of articles, newsletters, and websites. This manual process is highly inefficient, time-consuming, and prone to human bias. Furthermore, it is difficult to quickly separate high-quality, relevant information from noise, making it challenging to extract actionable insights and maintain a comprehensive understanding of rapidly evolving topics.

SOLUTION: To address this challenge, we developed the AI News Research Agent, an end-to-end multi-agent AI system that automates the entire information gathering and synthesis pipeline. By leveraging specialized AI agents to autonomously research, filter, summarize, and analyze web data, the system transforms scattered news into structured, insightful reports. An agentic approach is highly suitable here because the research process naturally decomposes into distinct, specialized tasks (searching, filtering, analyzing) that can be handled collaboratively in a pipeline.

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2. Task Decomposition & Specifications

The AI News Research Agent is designed as a multi-agent system where each agent is responsible for a specific sub-task. The workflow begins when the user provides a research topic and specifies the desired research depth.

Stage 1: Data Gathering (Research Agent): The Research Agent takes the user's topic and utilizes the Tavily Search tool to scour the web for the latest, most relevant news articles and publications. Its output is a raw dataset of web sources.

Stage 2: Noise Reduction (Filtering Agent): The Filtering Agent receives the raw search data. It evaluates the credibility and relevance of each source, removing duplicates, clickbait, and off-topic noise to curate a high-quality list of articles.

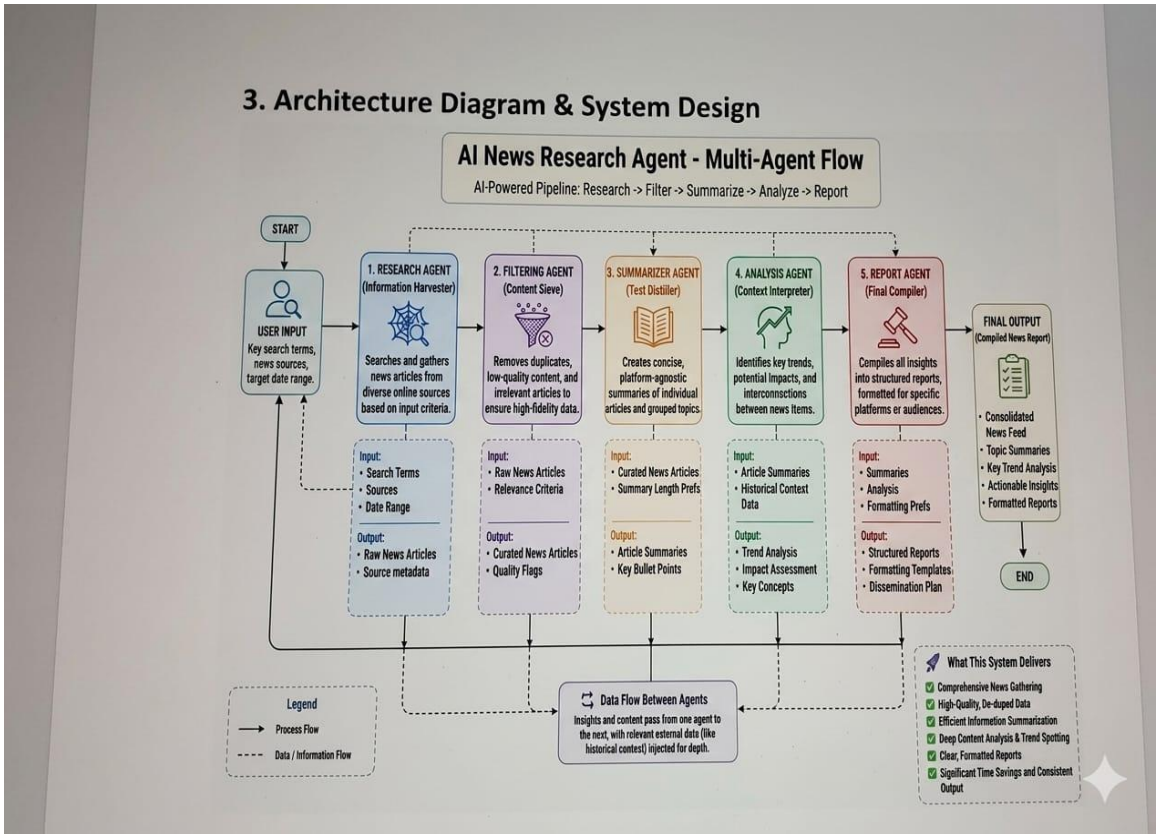
Stage 3: Information Extraction (Summarizer Agent): The Summarizer Agent processes the filtered articles, condensing lengthy texts into concise, factual summaries while retaining the core information.

Stage 4: Pattern Recognition (Analysis Agent): The Analysis Agent reviews the summaries to identify broader trends, implications, sentiment, and hidden patterns across the different sources.

Stage 5: Final Synthesis (Insight Report Agent): The Insight Report Agent combines all findings, summaries, and analyses into a beautifully formatted, comprehensive final report.

Decision points exist between each stage, ensuring that only high-quality, structured data is passed forward in the pipeline.

3. Architecture Diagram & System Design



4.Implementation

The project is implemented using Python as the primary programming language. The frontend interface is built using Streamlit to provide a premium, modern, and interactive web experience. The application integrates with Groq API as the primary LLM provider to ensure lightning-fast reasoning and text generation.Tool integration is achieved using Tavily Search API, enabling the Research Agent to retrieve real-time, accurate web-based insights. The system is highly modular, with separate Python files for each agent (e.g., research_agent.py, filtering_agent.py) to maintain clean architecture and separation of concerns.

5.LLM-as-Judge Design

The evaluation module uses an LLM-as-Judge framework to ensure the final generated report is of the highest quality before it is finalized. The judge follows a strict rubric-based scoring mechanism. The criteria include:

- **Factuality & Relevance: Does the report accurately reflect the retrieved sources without hallucinating?**
- **Bias Detection: Is the reporting objective, or does it lean heavily toward a specific narrative?**
- **Depth of Analysis: Are the insights superficial, or do they offer deep, actionable takeaways?**
- **Clarity & Readability: Is the report well-structured, easy to read, and professionally formatted? The LLM evaluates the drafted report against this rubric, assigning qualitative feedback and ensuring the output is trustworthy and polished.**

6. Deployment

The completed application is deployed on Streamlit Community Cloud to ensure public accessibility and ease of use. Deployment involved configuring secure environment variables (Streamlit Secrets) so that the live app can securely access the Groq and Tavily APIs without exposing the keys. The deployed link allows users, evaluators, and recruiters to access and test the live research pipeline in real-time.

7. Challenges Faced and Solutions

During development and deployment, several challenges were encountered:

- **Deployment Dependency Issues:** We faced build errors during deployment because Streamlit Cloud defaulted to Python 3.14, which lacked pre-built binary wheels for packages like Pydantic and Pillow. This was resolved by configuring the deployment settings to use a stable Python version (3.11/3.12).
- **Environment Variable Management:** Transitioning from a local .env file to cloud-based secrets required debugging API key configuration errors on the live server.
- **Agent Orchestration:** Ensuring that the output of one agent (e.g., raw search results) was perfectly formatted for the next agent (Filtering) required careful prompt engineering and JSON structuring. These challenges significantly improved our understanding of deployment environments, API integration, debugging workflows, and multi-agent pipeline orchestration.

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8. Business Value and Impact

The AI News Research Agent significantly reduces the manual hours traditionally required for deep-dive research. It ensures comprehensive, unbiased, and structured reporting. The system provides ready-to-use insights that can directly support financial analysts, market researchers, students, and content creators. By combining real-time tool usage (web search), sequential reasoning, and evaluation, the solution demonstrates immense real-world value, acting as a tireless digital research assistant

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9. Future Improvements

Future enhancements may include:

- **Automated Newsletters:** The ability to schedule the agent to run daily and email a customized news briefing.
- **Export Functionality:** Allowing users to download the final report as a PDF, Word Document, or Markdown file.
- **Multi-Source Integration:** Expanding the toolset to include specific database searches like arXiv for academic papers or Yahoo Finance for stock data.
- **Human-in-the-Loop:** A feature allowing users to manually review the filtered sources before the system generates the final summary.

10. Conclusion

The AI News Research Agent demonstrates how an agentic AI system can solve the practical problem of information overload through task decomposition, modular architecture, and real-time tool integration. By designing a structured multi-agent workflow, we transformed a manual, exhausting research process into an efficient, repeatable, and scalable system. This project reflects the practical implementation of agentic AI concepts and showcases our ability to design, build, deploy, and evaluate a complex, end-to-end AI application.